EXPERIMENT NO:6

USER INTERFACE DESIGN

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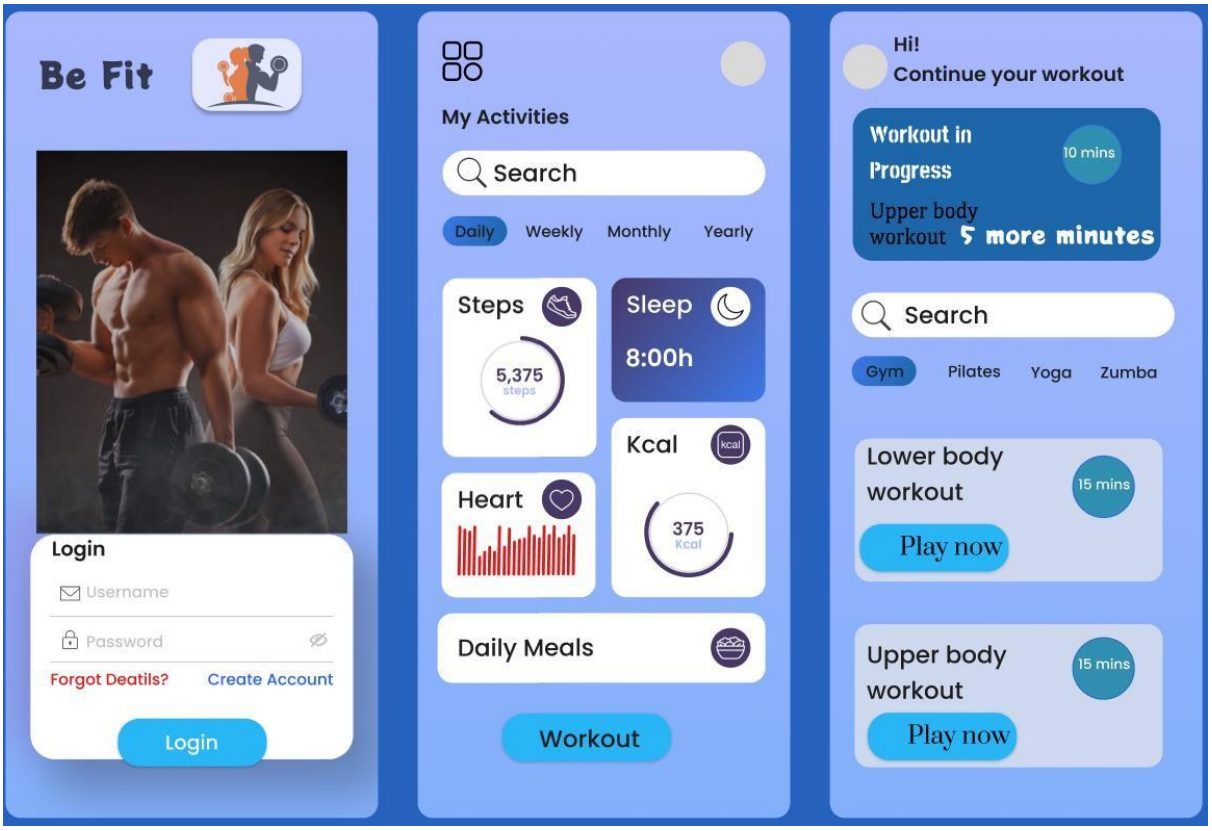

RAD for Fitness App

RAD (Rapid Application Development) is a software development methodology that emphasizes **quick development and iteration of prototypes** over strict planning. It is ideal for a fitness app where user needs evolve (e.g., workouts, tracking, personalization).

Key Characteristics of RAD

- Focus on **prototypes** instead of heavy documentation
- **User involvement** throughout development
- **Fast development cycles** with frequent updates
- **Reusable components** (e.g., workout templates, UI modules)
- Best suited for **UI/UX-intensive apps** like fitness tracking apps

Phase 1: Requirements Planning

This phase defines **project goals, system requirements, and user needs**.

1) Identify Key Features

Navigation:

- Home Dashboard
- Workout Plans
- Exercise Library
- Progress Tracking
- Nutrition Tracker
- Profile
- Settings

User Actions:

- Sign up / Login
- Browse workouts
- Start/track workouts
- Log exercises (sets, reps, time)
- Track calories and meals
- View progress (weight, stats)
- Set fitness goals
- Receive notifications/reminders

2) Create a Requirements Document

1. View personalized dashboard (daily activity, goals)
2. Browse workout categories (strength, cardio, yoga, etc.)
3. View exercises under each category
4. View detailed exercise instructions (video/demo)
5. Start and track a workout session
6. Log sets, reps, duration, calories burned
7. Track nutrition (meals, calories intake)

8. View progress charts (weight, performance)
9. Set and update fitness goals
10. Receive workout reminders and notifications

Phase 2: User Design

Designers and users collaborate to create **wireframes and prototypes**.

1) Tools Used

- Figma / Adobe XD (for UI/UX design)
- Prototyping tools for interaction design

2) Wireframes Designed

- **Home Dashboard:**
  Shows daily summary (steps, calories, active minutes, goals progress)
- **Workout Plans:**
  Displays recommended or selected workout routines
- **Exercise Library:**
  Lists exercises categorized by muscle group or type
- **Exercise Details:**
  Shows instructions, images/videos, and tracking options
- **Progress Tracker:**
  Displays charts for weight, workouts completed, calories burned
- **Nutrition Tracker:**
  Allows users to log meals and track calorie intake
- **Profile:**
  Displays user info, goals, achievements

3) Additional Screens

- **Workout Session Screen:**
  Tracks ongoing workout (timer, sets, reps)
- **Goal Setting Screen:**
  Users set fitness targets (weight loss, muscle gain)

- **Notifications:**
  Reminders for workouts, hydration, meals

4) Design Considerations

- Clean and motivating UI
- Easy navigation (minimal taps)
- Visual progress (graphs, badges)
- Accessibility (clear fonts, icons)

5) Interaction & Navigation

- Interactive buttons (Start Workout, Log Meal, etc.)
- Swipe between workout steps
- Real-time updates (progress tracking)
- Use of reusable UI components (cards, charts, buttons)